Reinforcement Learning-Based Power Management for University Computer Laboratories

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GitHub Repository: https://github.com/kunalsingh-iitj/rl-lab-power-management (Shared access: `@shubhambagwari`, `@divyaanshmertia`)

Abstract—University computer laboratories routinely leave workstation fleets fully powered on 24 hours a day to ensure instant availability for students, resulting in substantial electricity waste during overnight and off-peak periods. Conversely, aggressive static sleep policies degrade Quality of Service (QoS) via cold wake-up delays and login timeouts during scheduled class arrival surges. We formulate university lab power management across an $N = 20$ workstation fleet over a 24-hour horizon ($T = 96$ quarter-hour steps) as a finite Markov Decision Process (MDP) with $|\mathcal{S}| = 720$ discrete states and $|\mathcal{A}| = 3$ power modes (`SLEEP`, `IDLE`, `ACTIVE`). We train off-policy Tabular Q-Learning and on-policy SARSA controllers and benchmark them across 20 held-out random seeds (100 evaluation episodes per policy) against four baselines. Tabular Q-Learning achieves the highest mean cumulative reward (373.52 ± 26.64, outperforming `Always-Active` by +25.0% and `Always-Idle` by +159.9%), cuts daily electrical energy consumption by 30.7% (40.31 vs. 58.17 kWh/day) and Time-of-Use (ToU) electricity cost by 21.3% ($5.61 vs. $7.13/day), while preserving a 99.6% student service completion rate at 1.78 s average response latency.

1. Introduction & Problem Motivation

Academic computing facilities experience strongly non-stationary diurnal and weekly workload patterns. During scheduled morning (08:00–12:00) and afternoon (14:00–18:00) practical sessions on weekdays, workstation occupancy approaches capacity (16–20 active machines) with bursty student arrivals at slot boundaries. In contrast, during overnight hours (00:00–06:00), lunch breaks, and weekend nights, occupancy drops near zero.

Static power policies fail to resolve the multi-objective trade-off between energy efficiency and student QoS: (i) an `Always-Active` policy keeps unoccupied workstations in high-power P0 standby (110 W per machine), wasting nearly 18 kWh/day in idle power; (ii) an `Always-Sleep` policy places machines into Suspend-to-RAM S3 (8 W), but incurs 45 s wake-up latency, premature session suspensions, and 44.3% request abandonment; and (iii) an `Always-Idle` policy (45 W) handles light traffic well but suffers wake-up queue congestion during scheduled lab arrival surges. We address this sequential control problem using Reinforcement Learning (RL) [1, 3], enabling an autonomous agent to anticipate diurnal demand surges, exploit Time-of-Use (ToU) tariffs, and avoid hardware-degrading power-mode oscillations.

2. MDP Formulation

We model the 20-workstation laboratory over a 24-hour cycle divided into $T = 96$ discrete intervals of $\Delta t = 15$ min (0.25 h) as a discounted MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ with $\gamma = 0.96$.

2.1 State Space ($\mathcal{S}$) and Dimensionality

At step $t \in \{0, \ldots, 95\}$, the agent observes a 5-dimensional discrete state vector ($\dim(\mathcal{S}) = 5$):

$$s_t = (p_t,\, d_t,\, u_t,\, \delta_t,\, m_t) \in \mathcal{S} \tag{1}$$

- **Diurnal Period** $p_t = \lfloor h_t/3 \rfloor \in \{0, \ldots, 7\}$: 8 three-hour blocks spanning 00:00–23:59.
- **Day Type** $d_t \in \{0, 1\}$: 0 = Weekday (5/7 prob.), 1 = Weekend (2/7 prob.).
- **Occupancy Demand Bracket** $u_t \in \{0, 1, 2, 3, 4\}$: Discretized active user count $n_t \in \{0, \ldots, 20\}$ into 5 brackets: Bin 0 (0–1), Bin 1 (2–5), Bin 2 (6–10), Bin 3 (11–15), and Bin 4 (16–20 PCs).
- **Demand Momentum** $\delta_t \in \{0, 1, 2\}$: Step-over-step occupancy change $\Delta n_t = n_t - n_{t-1}$ categorized as Falling ($\Delta n_t \leq -2$), Stable ($|\Delta n_t| \leq 1$), or Rising ($\Delta n_t \geq 2$).
- **Current Standby Mode** $m_t \in \{0, 1, 2\}$: Operational mode of the standby pool from step $t-1$, required to preserve the Markov property under mode-transition latency and energy costs.

The exact cardinality of the discrete state space is:

$$|\mathcal{S}| = 8 \times 2 \times 5 \times 3 \times 3 = 720 \text{ states}, \tag{2}$$

mapped bijectively via mixed-radix encoding to $\{0, 1, \ldots, 719\}$.

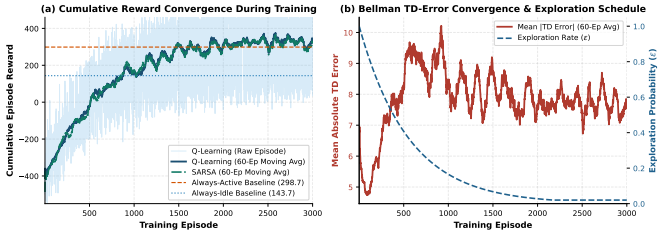


Figure 1: Training convergence over 3,000 episodes: (a) Cumulative episode reward for Q-Learning and SARSA surpassing static baselines; (b) Bellman TD-error stabilization alongside exponential ϵ-decay.

2.2 Action Space ($\mathcal{A}$)

The agent chooses a discrete target power mode $a_t \in \mathcal{A} = \{0, 1, 2\}$ ($|\mathcal{A}| = 3$) for the lab's unoccupied standby pool:

- $a_t = 0$ (`SLEEP`): Suspend-to-RAM S3 ($P_{\text{sleep}} = 8$ W/PC). Cold resume requires $\tau_{\text{sleep}} = 45.0$ s plus 2 s per queued arrival, causing 50% arrival timeout violations and a 20% premature suspend rate on paused sessions.
- $a_t = 1$ (`IDLE`): Low-power C-state with display off ($P_{\text{idle}} = 45$ W/PC, $\tau_{\text{idle}} = 5.5$ s). Serves up to $B = 3$ arrivals per step with zero violations; surge arrivals ($n_{\text{arr},t} > 3$) incur queue delay and a 40% timeout rate on excess surge requests.
- $a_t = 2$ (`ACTIVE`): High-performance P0 ready state ($P_{\text{active}} = 110$ W/PC; busy PCs draw $P_{\text{busy}} = 150$ W/PC). Serves all arrivals within capacity at $\tau_{\text{active}} = 1.5$ s latency (+4.0 s if transitioning directly from `SLEEP`) with zero service violations.

2.3 Environment Dynamics and Reward Function ($\mathcal{R}$)

At step t, existing users depart via $n_{\text{dep},t} \sim \text{Binomial}(n_t, p_{\text{dep}}(h_t, d_t))$ and new requests arrive via a non-homogeneous Poisson process $n_{\text{arr},t} = \min(N - n_{\text{ret},t}, \text{Poisson}(\lambda(h_t, q_t, d_t)))$, where weekday λ peaks at 5.2 arrivals/step during afternoon labs (14:00–18:00) and drops to 0.04 overnight. Step electrical energy (in kWh) combines active, standby, and transition overhead:

$$E_t = \frac{N_{\text{srv},t} P_{\text{busy}} + (N - N_{\text{srv},t}) P_{\text{stby}}(a_t)}{1000} \Delta t + E_{\text{sw}}(m_t, a_t) \tag{3}$$

where $E_{\text{sw}}(m_t, a_t) = 0.0035(N - N_{\text{srv},t})(1 + 0.4(|a_t - m_t| - 1))\mathbb{I}(a_t \neq m_t)$. Electricity cost is $C_t = c(h_t)E_t$ under a three-tier ToU tariff: $0.08/kWh off-peak (21:00–06:00), $0.12/kWh standard (06:00–12:00, 18:00–21:00), and $0.18/kWh peak (12:00–18:00).

The multi-objective reward function R_t is defined as:

$$\begin{aligned} R_t = {} & w_{\text{srv}} N_{\text{srv},t} - w_{\text{eng}} E_t - w_{\text{viol}} N_{\text{viol},t} \\ & - w_{\text{lat}} \left(\frac{L_t}{L_{\text{ref}}} \right) - w_{\text{sw}} \mathbb{I}(a_t \neq m_t) \end{aligned} \tag{4}$$

with weights $w_{\text{srv}} = 1.25$ per served student-step, $w_{\text{eng}} = 6.00$/kWh, $w_{\text{viol}} = 6.50$ per violation, $w_{\text{lat}} = 1.50$ ($L_{\text{ref}} = 10.0$ s), and $w_{\text{sw}} = 0.70$ per mode switch. When the lab is unoccupied with zero arrivals ($n_{\text{ret},t} = 0, n_{\text{arr},t} = 0$), user-experienced latency is $L_t = 0$ s.

Table 1: Quantitative Comparison Across 20 Independent Held-Out Evaluation Seeds ($20 \times 5 = 100$ Evaluation Episodes per Policy; Mean ± SD).

Policy	Cumulative Reward (↑)	Daily Energy (kWh ↓)	ToU Cost (USD/day ↓)	Served Requests/Day (↑)	Service Rate (↑)	QoS Violations/Day (↓)	Mean Latency (s ↓)	Mode Switches/Day
Q-Learning (Ours)	$\mathbf{373.52 \pm 26.64}$	40.31 ± 0.43	$\$5.61 \pm 0.06$	525.4 ± 18.8	$99.6\% \pm 0.1\%$	1.61 ± 0.56	1.78 ± 0.06	7.50 ± 1.13
SARSA (Ours)	367.60 ± 27.70	39.86 ± 0.45	$\$5.51 \pm 0.06$	524.7 ± 19.3	$99.5\% \pm 0.1\%$	2.08 ± 0.59	1.97 ± 0.07	10.40 ± 1.17
`Always-Active`	298.72 ± 26.82	58.17 ± 0.37	$\$7.13 \pm 0.05$	$\mathbf{529.8 \pm 19.6}$	$\mathbf{100.0\% \pm 0.0\%}$	$\mathbf{0.00 \pm 0.00}$	$\mathbf{0.96 \pm 0.02}$	1.00 ± 0.00
`Always-Idle`	143.74 ± 22.58	33.60 ± 0.56	$\$4.37 \pm 0.08$	457.3 ± 16.5	$95.5\% \pm 0.3\%$	22.70 ± 1.54	5.46 ± 0.18	$\mathbf{0.00 \pm 0.00}$
`Random`	-372.49 ± 40.00	37.90 ± 1.13	$\$4.75 \pm 0.14$	347.4 ± 16.4	$86.2\% \pm 0.7\%$	54.89 ± 3.55	12.34 ± 0.46	64.07 ± 2.02
`Always-Sleep`	-1029.45 ± 24.54	$\mathbf{9.21 \pm 0.39}$	$\mathbf{\$1.27 \pm 0.05}$	149.3 ± 7.5	$55.7\% \pm 0.8\%$	114.33 ± 3.87	28.96 ± 0.57	1.00 ± 0.00

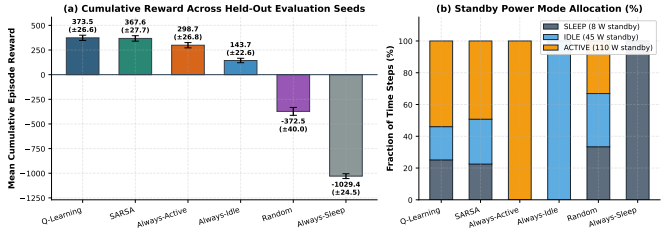


Figure 2: Held-out evaluation across 20 seeds (100 episodes/policy): (a) Mean cumulative episode reward (±1 SD); (b) Percentage of steps spent in `SLEEP`, `IDLE`, and `ACTIVE` modes.

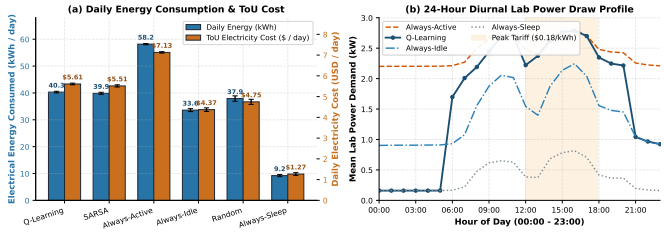


Figure 3: Energy and cost analysis: (a) Daily energy consumption (kWh) and ToU cost ($); (b) 24-hour diurnal power profile showing Q-Learning sleeping overnight and scaling up for daytime labs.

3. RL Algorithms & Baseline Policies

Because the state-action space has $|\mathcal{S}| \times |\mathcal{A}| = 720 \times 3 = 2,160$ entries, **Tabular Q-Learning** [2] is ideally suited: it avoids function-approximation divergence, guarantees convergence to $Q^*(s, a)$, and yields an inspectable policy table. At each transition (s_t, a_t, R_t, s_{t+1}), the off-policy TD(0) update is:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha_k \Big[R_t + \gamma(1 - d_{\text{done}}) \max_{a' \in \mathcal{A}} Q(s_{t+1}, a') - Q(s_t, a_t) \Big] \tag{5}$$

We also train on-policy **Tabular SARSA**, replacing $\max_{a'} Q(s_{t+1}, a')$ with $Q(s_{t+1}, a_{t+1})$ under the behavior policy. Both agents train for $K = 3,000$ episodes (288,000 steps, training seed 42) using ϵ-greedy exploration with unvisited-action prioritization ($\epsilon_0 = 1.0 \to \epsilon_{\min} = 0.02$, decay $\lambda_\epsilon = 0.9982$) and learning rate decay ($\alpha_0 = 0.15 \to \alpha_{\min} = 0.015$, $\lambda_\alpha = 0.9992$). We compare against four baselines: (1) `Always-Active` ($a_t = 2$), (2) `Always-Idle` ($a_t = 1$), (3) `Always-Sleep` ($a_t = 0$), and (4) `Random` ($a_t \sim \mathcal{U}\{0, 1, 2\}$).

4. Experimental Evaluation & Results

To prevent training–evaluation leakage, evaluation is conducted on 20 strictly disjoint held-out random seeds (`1000..1019`), each running 5 episodes (4 weekdays + 1 weekend, totaling 100 episodes per policy) with greedy action selection ($\epsilon = 0$).

Training Convergence: Fig. 1(a) shows that both Q-Learning and SARSA rapidly improve from −362.9 initial episode reward (under random exploration) to surpass `Always-Idle` (143.7) by episode 900 and `Always-Active` (298.7) by episode 1,450, converging smoothly as $\epsilon \to 0.02$ and mean TD error stabilizes (Fig. 1(b)). During training, 404/720 physically reachable diurnal states are visited across 288,000 updates.

Quantitative Comparison: Table 1 and Figs. 2–4 summarize held-out performance across all metrics:

1. **Cumulative Reward: Q-Learning** achieves the highest reward of $\mathbf{373.52 \pm 26.64}$, outperforming `Always-Active` (298.72) by $+\mathbf{25.0}\%$ (+74.80 points) and `Always-Idle` (143.74) by $+\mathbf{159.9}\%$. On-policy **SARSA** achieves 367.60 ± 27.70, slightly more conservative due to accounting for exploratory risk during TD backups.

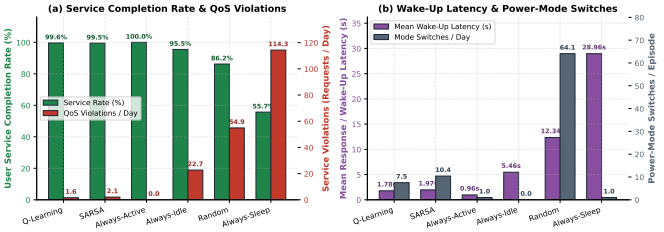


Figure 4: Service quality metrics: (a) Student service completion rate (%) and daily QoS violations; (b) Mean wake-up latency (s) and daily mode-switching frequency across policies.

2. **Energy & Cost Reduction:** Compared to `Always-Active`, Q-Learning reduces daily energy consumption by **30.7**% (40.31 vs. 58.17 kWh/day, saving 17.86 kWh/day or 6,519 kWh/year per 20-PC lab) and lowers ToU electricity cost by **21.3**% ($5.61 vs. $7.13/day). As shown in Fig. 3(b), Q-Learning drops lab power to 0.16 kW (`SLEEP`) during 00:00–06:00, tracks active lab demand during 08:00–20:00, and ramps down to 0.95 kW (`IDLE`) after 21:00.
3. **QoS, Violations & Latency:** Q-Learning serves 525.4 requests/day (**99.6**% service rate) with only 1.61 violations/day and 1.78 s mean latency—eliminating 92.9% of the violations incurred by `Always-Idle` (22.70/day, 5.46 s) and 98.6% of those under `Always-Sleep` (114.33/day, 28.96 s). Moreover, Q-Learning executes only 7.50 mode switches/day (vs. 64.07 for `Random`).

5. Discussion, Limitations & Conclusion

Fig. 5 and Fig. 2(b) reveal the learned multi-regime strategy $\pi^*(s)$: Q-Learning selects `SLEEP` (25.2% of steps) during empty overnight periods (00:00–06:00), `IDLE` (20.8% of steps) during low-load midday/evening blocks and weekend study hours, and proactively switches to `ACTIVE` (54.0% of steps) during weekday lab periods to absorb bursty student arrivals without wake-up timeouts.

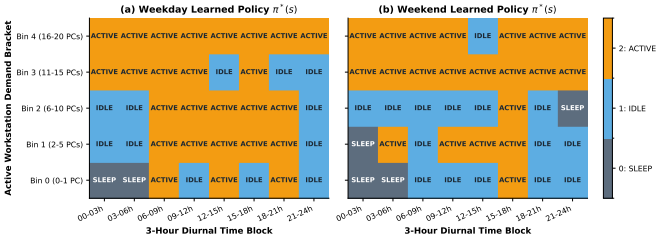


Figure 5: Learned greedy Q-Learning policy $\pi^*(s)$ across diurnal blocks and occupancy brackets for (a) Weekdays and (b) Weekends.

Limitations & Future Work: Our current environment applies a unified standby mode across all unoccupied PCs and assumes homogeneous workstation hardware. Extending the action space to fine-grained cluster sizing (maintaining a dynamic warm-idle buffer of $k \in \{0, \ldots, N\}$ PCs) using Deep Q-Networks (DQN) [6] conditioned on academic calendar deadlines is a promising direction.

References

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